

Lesson 08

Consolidation of Soil



Topics for the day;

- Compressibility of soil
- Consolidation & Consolidation Settlement
- Oedometer Test
- Normally consolidated & over consolidated soils
- Soft ground treatment

Compressibility of Soil

When a soil mass is subjected to a compressive force, its volume decreases. This property of the soil is known as the **compressibility** of soil.

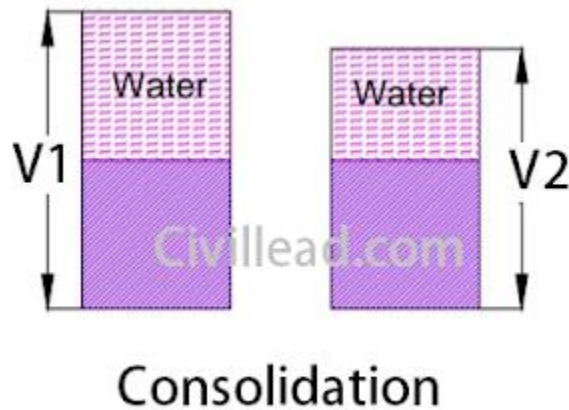
The compression of soil can occur due to;

- compression of solid particles and water in the voids,
- expulsion of air in the voids (**compaction**),
- expulsion of water in the voids (**consolidation**).

Consolidation & Consolidation settlement of soil

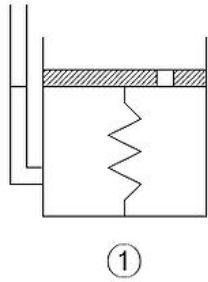
The gradual reduction in soil volume of a saturated soil due to the removal of water from the soil voids caused by loading is defined as **consolidation**.

Vertical displacement due to the volume change is called the **consolidation settlement**.

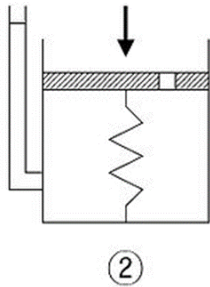


Spring analogy

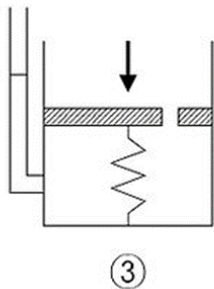
The process of consolidation is explained with a system composed of a spring, a container with a hole in its cover, and water.



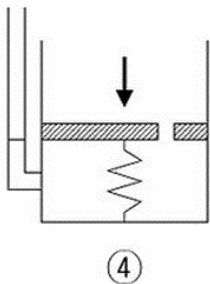
The container is filled with water, and the hole is closed. (fully saturated soil)



A load is applied, the hole is still unopened. Only the water resists the applied load. (development of excess pore water pressure)

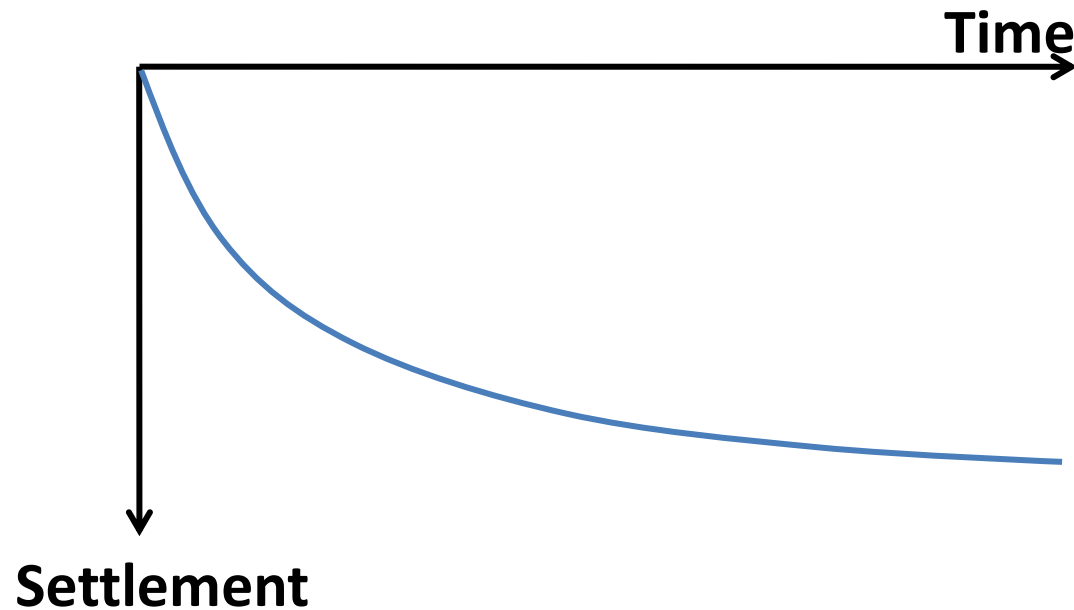


The hole is opened, water starts to drain out and the spring shortens. (drainage due to excess pore water pressure) Consolidation started.



After some time, the drainage of water stops. Now, the spring alone resists the applied load. (end of consolidation)

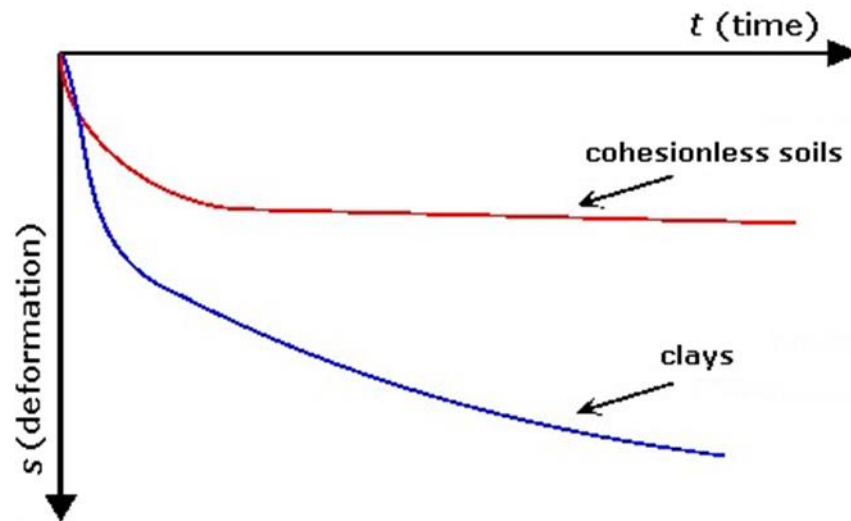
Consolidation process takes time, and the rate of settlement decreases over time.



The consolidation settlement which occurs in a given time depends on the;

- permeability of the soil
- length of the drainage path
- compressibility of the soil
- boundary conditions

Settlement of sands and clays



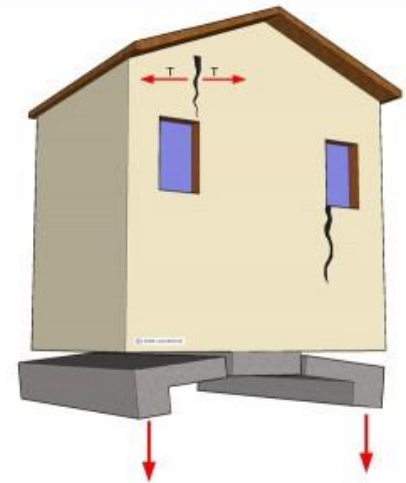
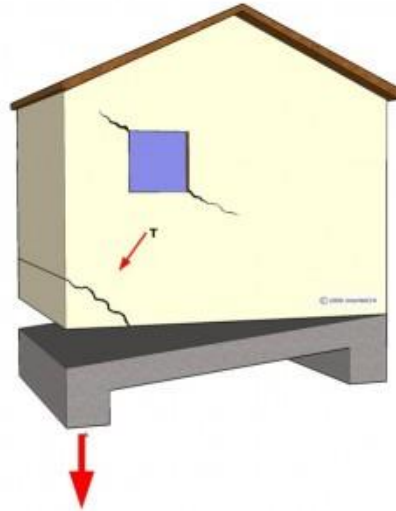
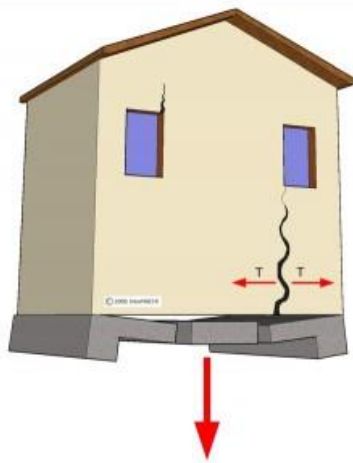
Sands and gravels (cohesionless soils)

Increase in pore pressures are dissipated rapidly due to high permeability. Therefore any volume change/settlement resulting from a change in loading occurs immediately.

Clays (cohesive soils)

Slow seepage occurs and the excess pore pressures dissipate slowly, consolidation settlement occurs.

Problems related to consolidation settlement



Problems related to consolidation settlement



Hence in civil engineering, it is important to know;

- the total consolidation settlement in mm/m
- the rate of consolidation settlement in m/mm/year

By knowing these parameters, the designer would be able to set necessary precautions and design considerations to keep the settlement to tolerable limits.

Consolidation settlement is measured using two parameters;

- Coefficient of Volume Compressibility(m_v)
- Coefficient of Consolidation (C_v)

Coefficient of volume compressibility (m_v)

It is defined as the change in volumetric strain per unit change in pressure /effective stress in one-dimensional compression. (units m^2 / MN)

$$m_v = \frac{\frac{\Delta V}{V}}{dp}$$

Once the coefficient of volume compressibility (m_v) of a soil is obtained, the total settlement of a soil layer of thickness H can be calculated.

$$\text{Total settlement} = m_v dp H$$

dp is the increase in pressure/effective stress

Coefficient of consolidation - C_v

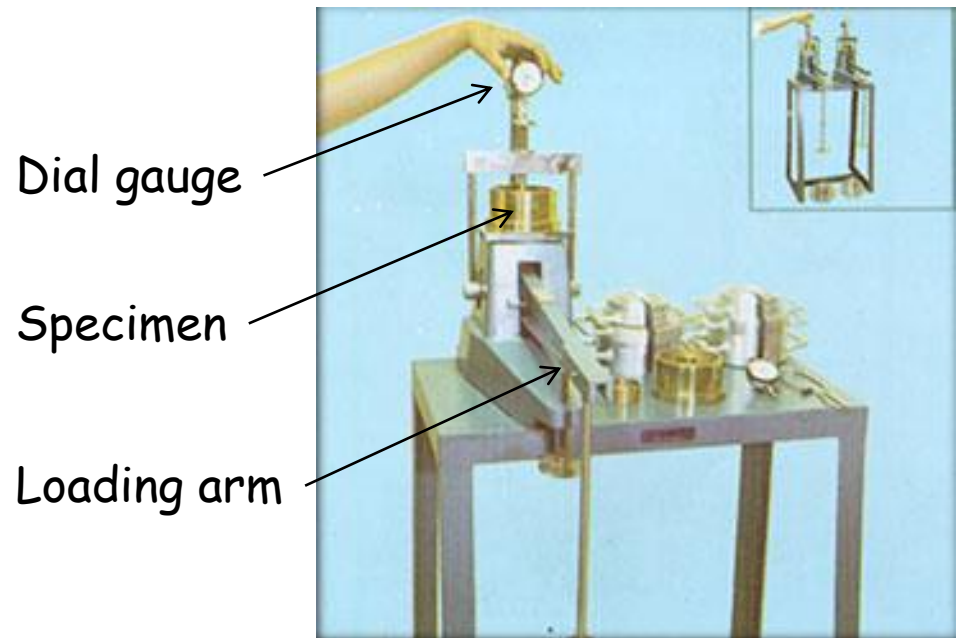
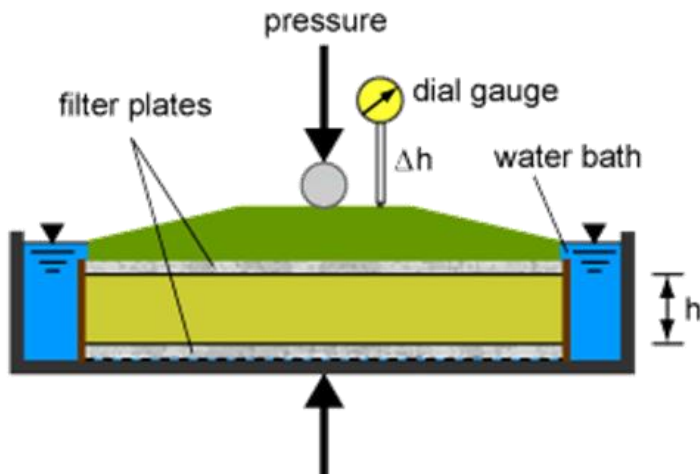
This value predicts the time rate of settlement of soil in the field. (units; mm^2/min or m^2/year)

Coefficient of volume compressibility (m_v) and coefficient of consolidation (C_v) are determined by a one dimensional consolidation test called the Oedometer test.

Oedometer Test

The one-dimensional compression and swelling characteristics of a soil may be measured in the laboratory using the **oedometer test**.

Apparatus



Procedure

A cylindrical specimen of soil enclosed in a metal ring is subjected to a series of increasing static loads, while changes in thickness are recorded against time.

Vertical static load increments are applied at regular time intervals (e.g. 12, 24, 48 hr)

The load is doubled with each increment up to the required maximum (e.g. 25, 50, 100, 200, 400, 800 kPa)

After full consolidation is reached under the final load, the loads are removed and the specimen allowed to swell.

After which the specimen is removed and its thickness and water content determined.

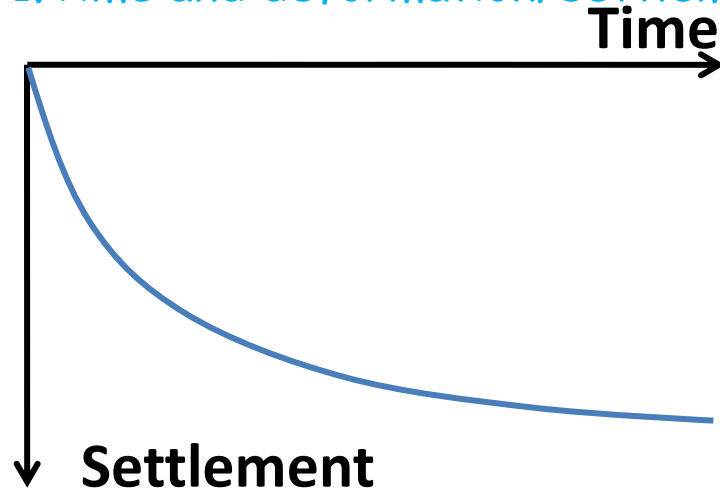
The test is fully detailed in BS 1377.

Observations & Results

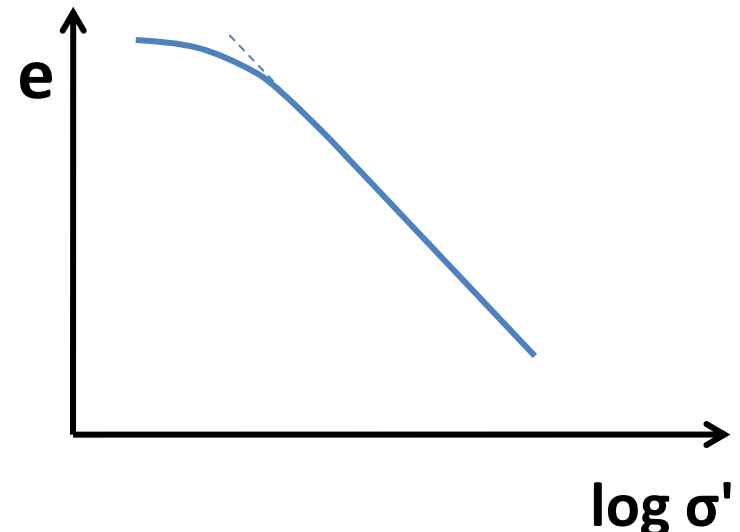
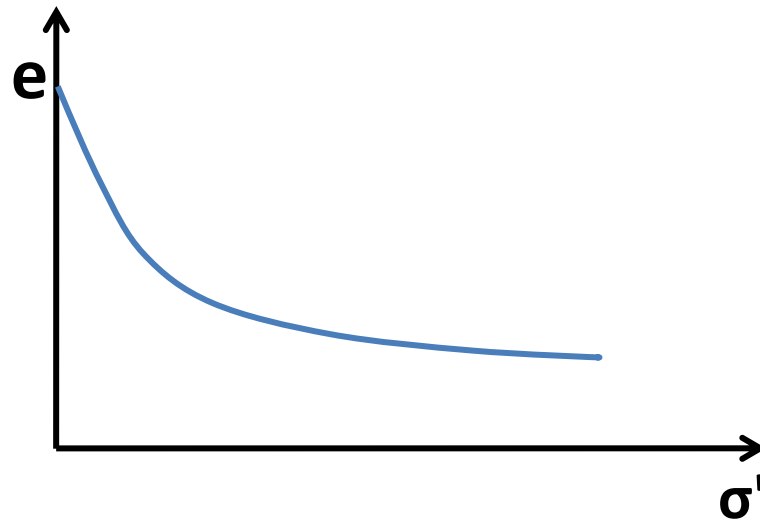
1. From the changes in thickness at the end of each load stage the **compressibility** of the soil may be observed, and parameters such as **Compression Index (C_c)** and **Coefficient of Volume Compressibility (m_v)** can be calculated.
2. From the changes in thickness recorded against time during a load stage the **rate of consolidation** may be observed and the **coefficient of consolidation (c_v)** calculated.

Typical Test Graphs

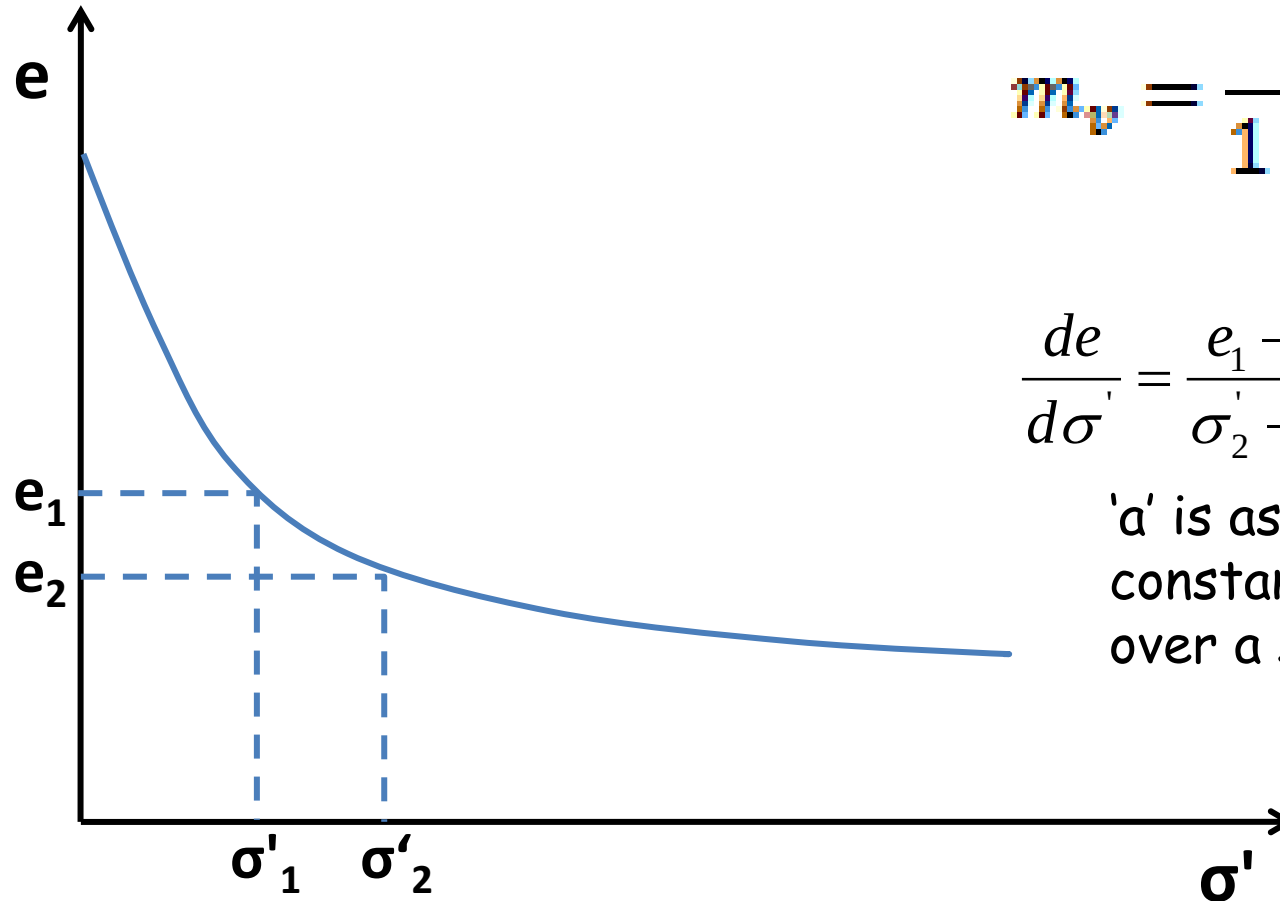
1. Time and deformation/settlement



2. Effective stress and void ratio



1. Coefficient of Volume Compressibility(m_v)



$$m_v = \frac{a}{1 + e_1}$$

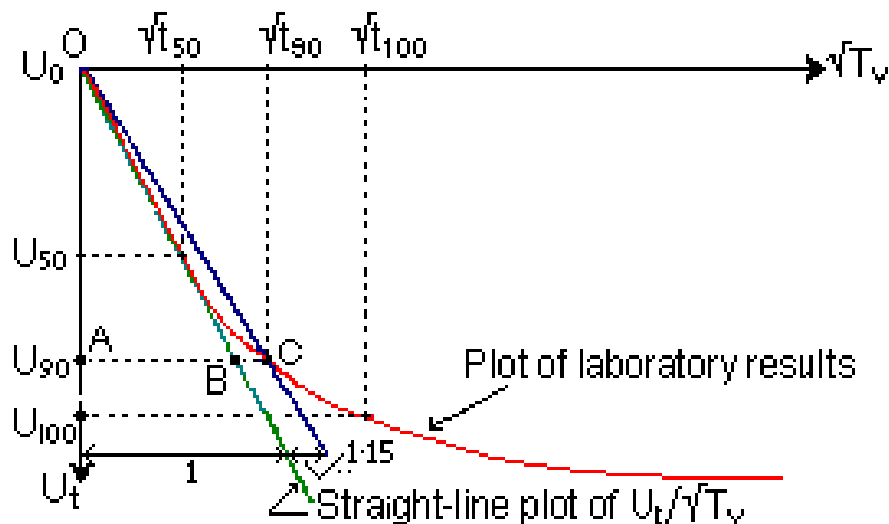
$$\frac{de}{d\sigma'} = \frac{e_1 - e_2}{\sigma'_2 - \sigma'_1} = a$$

'a' is assumed to be a constant over a stress range

2. Coefficient of Consolidation (c_v)

The procedure involves plotting thickness changes (i.e. settlement) against a suitable function of time (square root/log)

• The Root-Time method

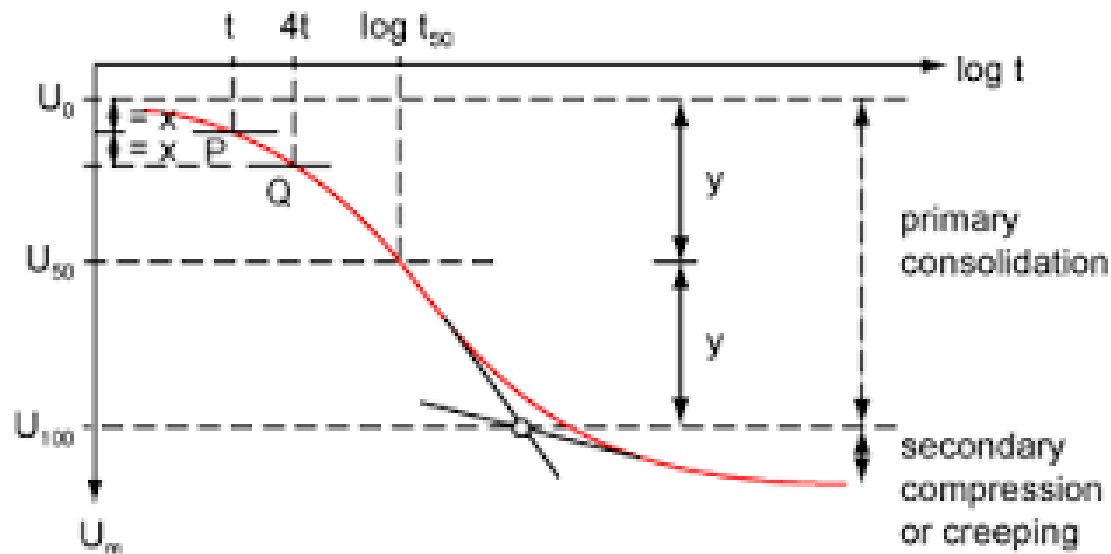


$$C_v = 0.848 \frac{d^2}{t_{90}}$$

'd' is the length of the drainage path

C_v is determined by estimating the time at 90% consolidation (t_{90})

- The Log-Time method



$$C_v = 0.197 \frac{d^2}{t_{50}}$$

'd' is the length of the drainage path

C_v , is determined by estimating the time at 50% consolidation (t_{50})

Normally consolidated and over consolidated soils

Normally consolidated soil

If the soil is such that it has never experienced pressure of the current magnitude in its entire history, it is called a normally consolidated soil.

Over consolidated soil

The soil is called pre-consolidated (or over-consolidated) if at any time in history, it has been subjected to a pressure equal to or greater than the current pressure applied to it.

In case of normally consolidated soils, the consolidation will be greater than that for a pre-consolidated soil.

That is because the pre-consolidated soil has previously experienced greater or equal pressure and has undergone at least some consolidation under that pressure.

So a pre-consolidated soil is preferred over a normally consolidated soil as a construction site.

Soft Ground Treatment

If the ground does not have enough bearing capacity to hold the structure on it, it is called a soft ground.

Types of soft ground

- Saturated clay ground
- Sandy ground with loose structure
- Reclaimed area on river/sea
- Reclaimed area by waste

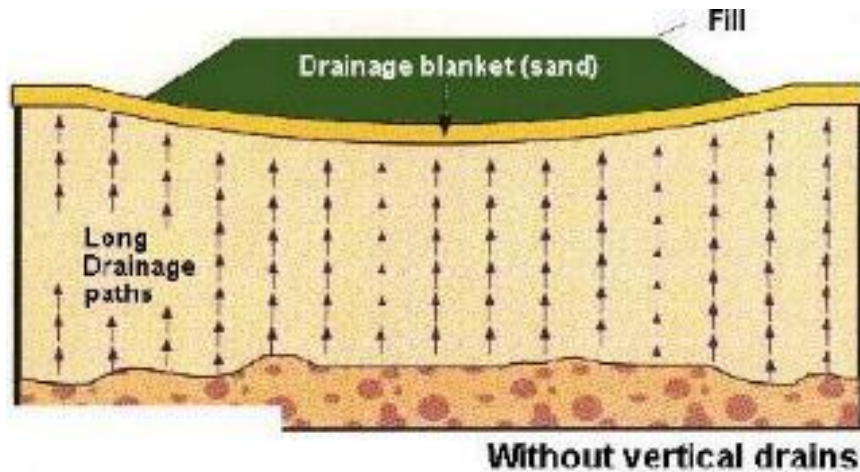
Methods of soft ground treatment

Construction work increases the load on soft soil layers, causing widespread settling. Waiting for the natural settlement of these lands is time consuming. To accelerate the settlement process, specialized solutions have been developed.

Drainage Methods

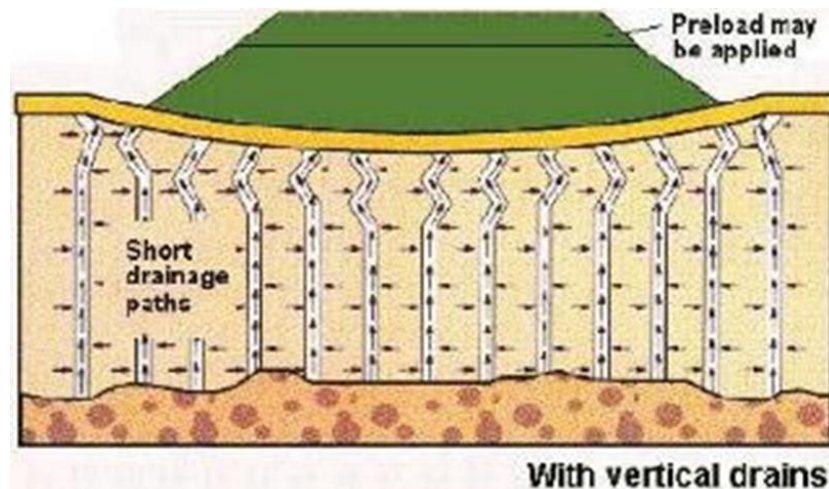
Soft, saturated clays take months or years to consolidate under an added load while excess pore pressures dissipate and the effective stress in the soil increases. Pre-compression and Vertical drains are two methods used to treat saturated clay soils by accelerating consolidation settlement.

1. Pre - compression/loading



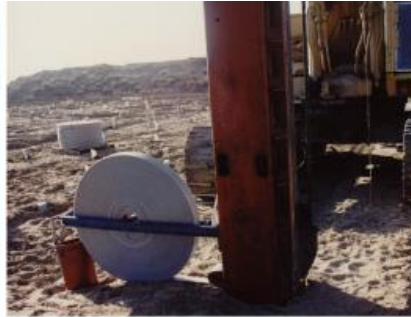
preloading technique using a surcharge embankment.

2. Vertical drains



vertical drains provide much shorter and highly permeable drainage paths for the dissipation of excess pore water pressure.

Installation of vertical drains



Summary;

- The compression of saturated soil under loading due to expulsion of water from the voids is termed as consolidation.
- Vertical displacement due to the volume change is called the consolidation settlement.
- Consolidation settlement is measured using two parameters;
Coefficient of Volume Compressibility(m_v)
Coefficient of Consolidation (C_v)
- These parameters are measured in the laboratory using the oedometer test.
- Consolidation of normally consolidated soils, will be greater than that for a pre-consolidated soil.
- Soft ground can be treated by accelerating the settlement process.